NCAI
NATIONAL CENTRE OF
ARTIFICIAL INTELLIGENCE

RAI-853: Advanced Robotics with ROS

Lec 1b: ROS Installation

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Outline

- Which ROS 2 Distribution to Choose?
- Installing the OS (Ubuntu)
- Installing Ubuntu 24.04 on a VM
- Installing ROS 2 on Ubuntu
- Installing Tools

Outline

- **Which ROS 2 Distribution to Choose?**
- Installing the OS (Ubuntu)
- Installing Ubuntu 24.04 on a VM
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Introduction

- Before using ROS 2, we need to install it and set it up. Doing this is not as trivial as just downloading and installing a basic program.

- There are several ROS 2 versions (called distributions), and we need to choose which one is the most appropriate.

- We also need to pick an Ubuntu version as ROS and Ubuntu distributions are closely linked.

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Jazzy Jalisco	May 23, 2024		May 2029
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Introduction

- Once you know which ROS/Ubuntu combination you need, you have to install the corresponding Ubuntu operating system (OS).

- Although being familiar with Linux is a prerequisite for this course, we will still do a recap on how to install Ubuntu on a virtual machine (VM), just in case, so you won't be lost and can continue with this course.

- Then, we will install ROS 2 on Ubuntu, set it up in our environment, and install additional tools that will allow you to have a better development experience.

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Which ROS 2 Distribution to Choose?

- Before we install ROS 2, it's important to know which distribution you need to use.

- ROS 2 is a project in continuous development, constantly receiving new features or improvements to existing ones. A **distribution** is simply a freeze in the development at some given point to create a stable release.

- With this, we can be sure that the core packages for one given distribution will not have any breaking changes. Without distributions, it would be impossible to have a stable system, and we would need to update our code constantly.

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Which ROS 2 Distribution to Choose?

- Every year, a new ROS 2 distribution is released on 23rd May. This day corresponds to World Turtle Day.

- As you will be able to observe, all ROS distributions have a turtle as a logo; there's a mobile robot platform named TurtleBot and even a 2D educational tool named Turtlesim.

- This is based on a reference to an educational programming language from 1967 named Logo, which included a feature to move some kind of turtle robot on the screen.

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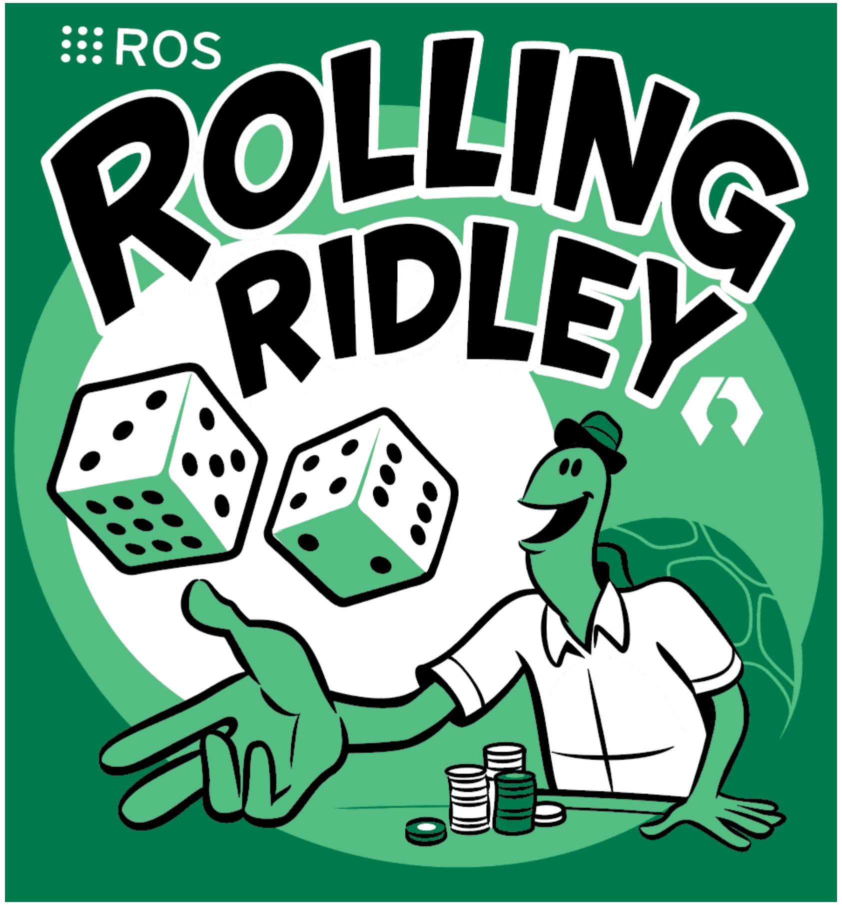
Which ROS 2 Distribution to Choose?

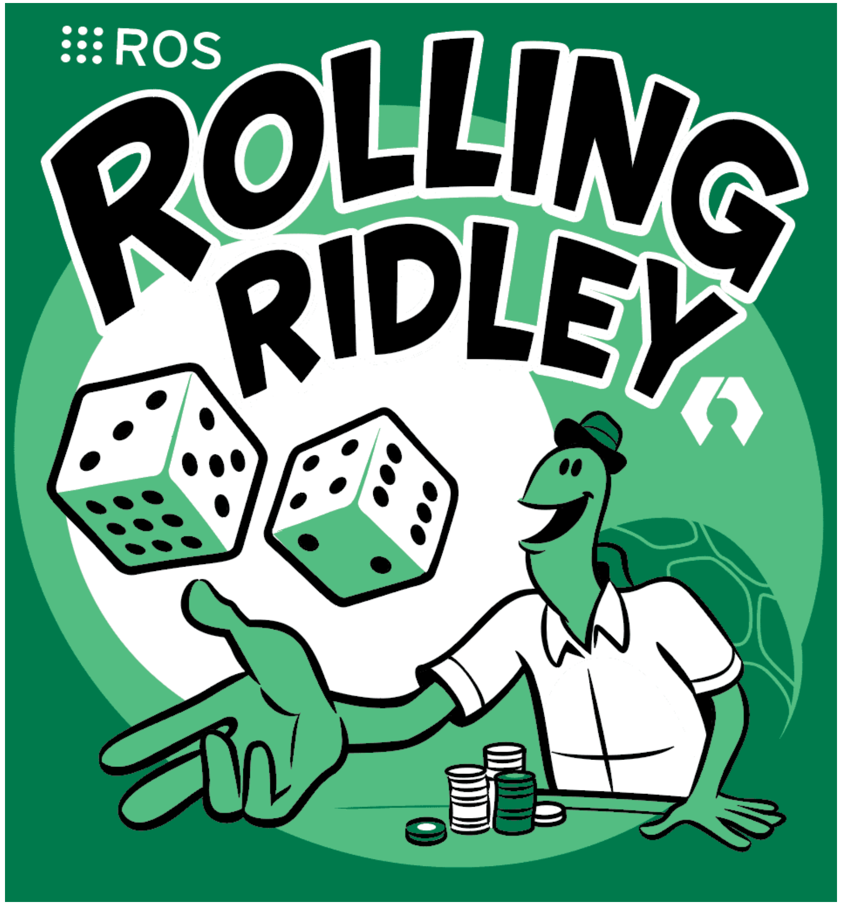
- On top of all the other distributions, there is another one that exists in parallel: **ROS Rolling**.
- This distribution is a bit special and is the distribution where all new developments are made.
- To make an analogy with Git and versioned systems, it's just like having a development branch and using this branch to release stable versions once a year.



Which ROS 2 Distribution to Choose?

- Thus, ROS Rolling is not a stable distribution, and we don't recommend using it for learning or to release a product.

- This is a distribution you can use if you want to test brand-new features before they are officially released into the next stable distribution—or if you want to contribute to the ROS code.



LTS and Non-LTS Distributions

- If you look a bit closer at the ROS homepage, you'll see that some distributions are supported for 5 years, others for 1.5 years (this only applies after 2022).
- You can see this by comparing the release date with the end-of-life (EOL) date.
- Currently supported distributions also have a green background on the screen, so we can easily spot them.

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LTS and Non-LTS Distributions

- When a distribution reaches its EOL date, it just means that it will not receive official support and package updates anymore.
- This doesn't mean you can't use it (in fact, lots of companies are still using legacy versions from 5 years ago or more), but you won't get any updates.
- The first official ROS 2 distribution was ROS Ardent, released in December 2017. After that, the first few distributions were still not quite complete, and the development team preferred to release shorter distributions so that the development could go faster.

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LTS and Non-LTS Distributions

- ROS Humble was the first Long-Term Support (LTS) release supported for 5 years (2022-2027).
- ROS Jazzy is also an LTS version, with official support from 2024 to 2029.
- From this, we can expect that every 2 years (even number: 2024, 2026, 2028, and so on), a new LTS distribution will be released in May and supported for 5 years.
- A few LTS distributions can coexist. In 2026, for example, with the release of the ROS L distribution, we will be able to use ROS Humble and ROS Jazzy as well.

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LTS and Non-LTS Distributions

- Then, you have non-LTS distributions. Those are released on odd years (2023, 2025, 2027, and so on) and are supported for 1.5 years only.

- Those distributions are released just so you can have access to new development in a somehow stable release without having to wait for 2 years.

- However, due to the short lifespan of non-LTS distributions and the fact that they will probably be less stable (and less supported), it is best not to use them if your goal is learning, teaching, or using ROS for a commercial application.

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How To Choose a ROS Distribution?

- What we recommend you to use the latest available LTS distribution.
- However, you are advised not to use an LTS distribution just after it's been released because it can still contain some bugs and problems.
- Also, some of the plugins and other community packages you need might not have been ported yet.
- Generally, if you want to work with a stable system, it's sometimes best not to stay too close to the new and shiny technology and wait a bit.

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How To Choose a ROS Distribution?

- In this course, we will use **ROS Jazzy**, which was released in May 2024.
- You can learn with one distribution and then start a project with another one.
- If you have a project or job that requires you to use a different ROS 2 distribution, you can still start to learn with ROS Jazzy.
- The gap between distributions is very small, especially for the core functionalities.

Outline

- Which ROS 2 Distribution to Choose?
- **Installing the OS (Ubuntu)**
- Installing Ubuntu 24.04 on a VM
- Installing ROS 2 on Ubuntu
- Installing Tools

Installing the OS (Ubuntu)

- ROS 2 works on three OSs: Ubuntu, Windows, and macOS.
- Although Ubuntu and Windows get Tier 1 support, macOS has only Tier 3 support, meaning “best effort,” not “fully tested.”
- This means that using macOS for ROS 2 is not necessarily the best choice for learning (if you’re an Apple user). We’re left with Windows or Ubuntu.

Installing the OS (Ubuntu)

- From teaching experience, we have seen that even if ROS could work well on Windows, it's not easy to install and use it correctly.

- When you're learning ROS, you want a smooth experience, and you want to spend time learning the features, not fixing the configuration.

- Hence, the **best overall option is to use Ubuntu**. If you don't have Ubuntu and are using Windows/macOS, you can either install Ubuntu natively as a dual boot on your computer or use a **Virtual Machine(VM).**

Relationship between ROS 2 and Ubuntu

- If we go to the Jazzy release page, we'll see that ROS Jazzy is supported only on Ubuntu 24.04.

- There is a close relationship between ROS and Ubuntu distributions. This relationship is quite simple.

- For every new Ubuntu LTS distribution (every 2 years on an even number), there is a new ROS 2 LTS distribution.

Relationship between ROS 2 and Ubuntu

- Ubuntu 22.04: ROS Humble

  Ubuntu 24.04: ROS Jazzy

  Ubuntu 26.04: ROS L

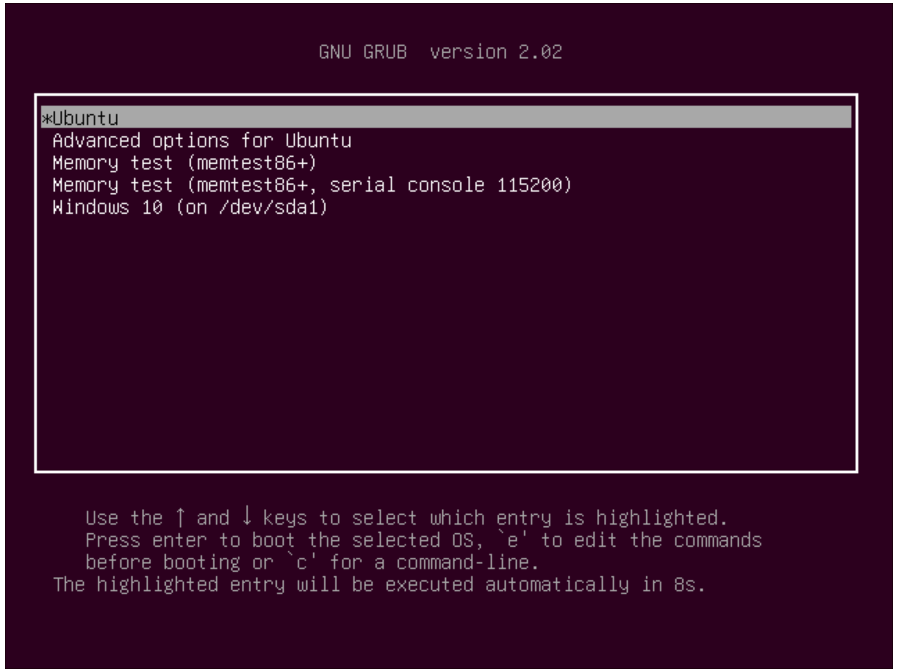
- It's important to use the correct combination. Thus, before installing ROS Jazzy, the first thing you must do is make sure you have Ubuntu 24.04 installed on your computer.

- If you happen to have an older version, we strongly encourage you to upgrade or simply install Ubuntu 24.04 from scratch.

Installing Ubuntu 24.04 Dual Boot

- The best option is to have Ubuntu installed natively on your computer. This will allow you to follow this course and then go further without any problems.

- Here are the high-level important steps you have to follow:
  1. Free some space on your disk so that you can create a new partition. We recommend a minimum of 70 GB, more if possible.
  2. Download an Ubuntu .iso file from the official Ubuntu website (Ubuntu 24.04 LTS).
  3. Flash this image on an SD card or USB key (with a tool such as **Balena Etcher**).

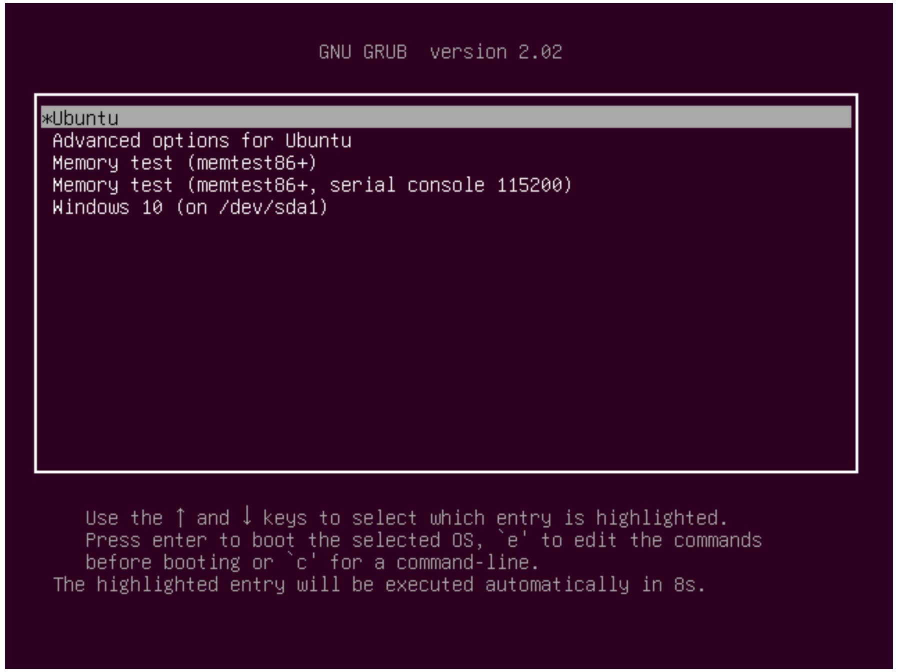


Installing Ubuntu 24.04 Dual Boot

4. Reboot your computer and choose to boot from the external device.

5. Follow the installation instructions. Important: When asked how you want to install, select Alongside Windows, for example—don't erase all your disk.

6. Complete the installation. Now, when you boot your computer, you should get a menu where you can select whether you want to start Ubuntu or Windows.



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Installing Ubuntu 24.04 on a VM

- If you can’t install Ubuntu as a dual boot (due to technical restrictions, lack of admin rights on the computer, or something else), or if you want to get started quickly with not too much effort, just for the sake of learning ROS, then you might wish to use a VM.

- A VM is quite easy to set up and is useful for teaching and learning purposes.

- Please note that 3D simulation and Gazebo will probably not work well on a VM. You can still use a VM for most of this course and set up a dual boot later when we will reach these topics.

Installing Ubuntu 24.04 on a VM

Step 1 – Downloading the Ubuntu .iso File

- Download the Ubuntu 24.04 .iso file. Note that, just like ROS, Ubuntu distributions also have a name. For Ubuntu 24.04, the name is Ubuntu Noble Numbat.

- Click on 64-bit PC (AMD64) desktop image. The file should be 5 to 6 GB, so make sure you have a good internet connection before downloading it.

Installing Ubuntu 24.04 on a VM

Step 2 – Installing VirtualBox

- You can start Step 2 while the Ubuntu .iso file is being downloaded.

- Two popular VM managers have a free version: VMware Workstation and VirtualBox.

- Both would work, but here, we will focus on VirtualBox as it's slightly easier to use.

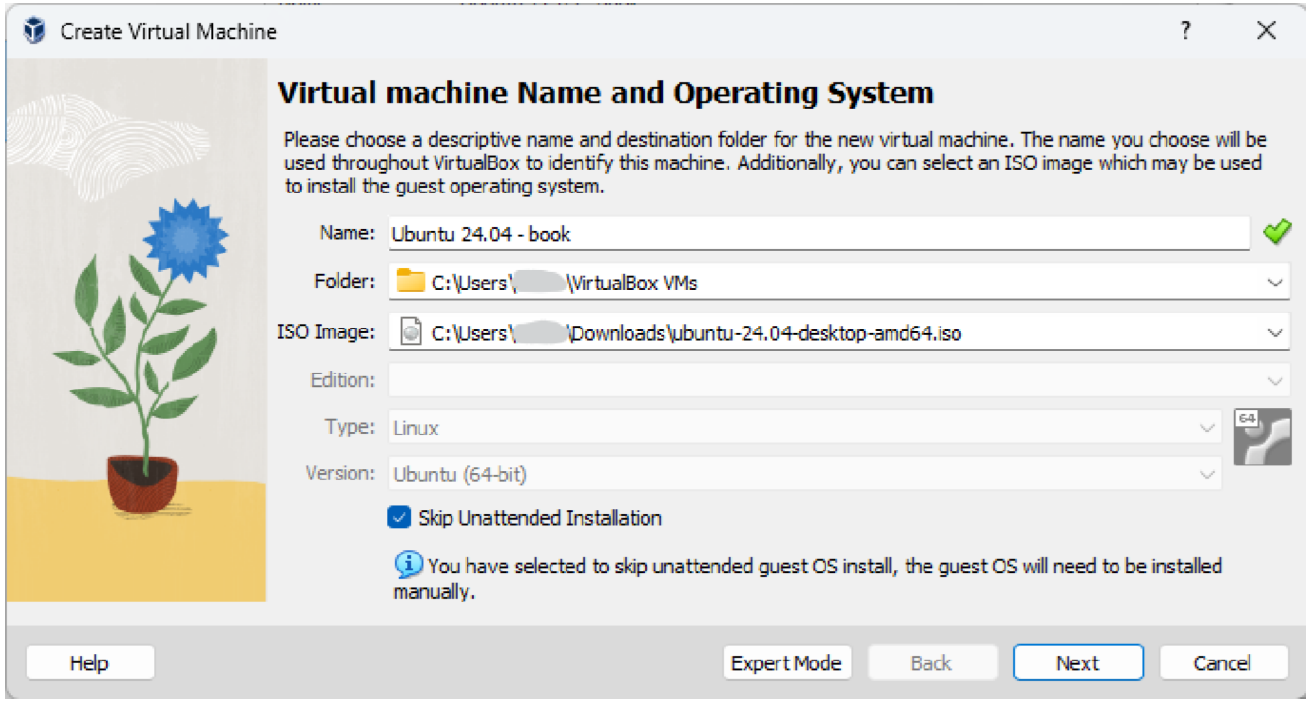
Installing Ubuntu 24.04 on a VM

- Go to the download page of the official VirtualBox website. Under VirtualBox Platform Packages, select the current OS you are running.
- If you want to install VirtualBox on Windows, for example, choose Windows hosts.
- Download the installer, then install VirtualBox like any other software.

Installing Ubuntu 24.04 on a VM

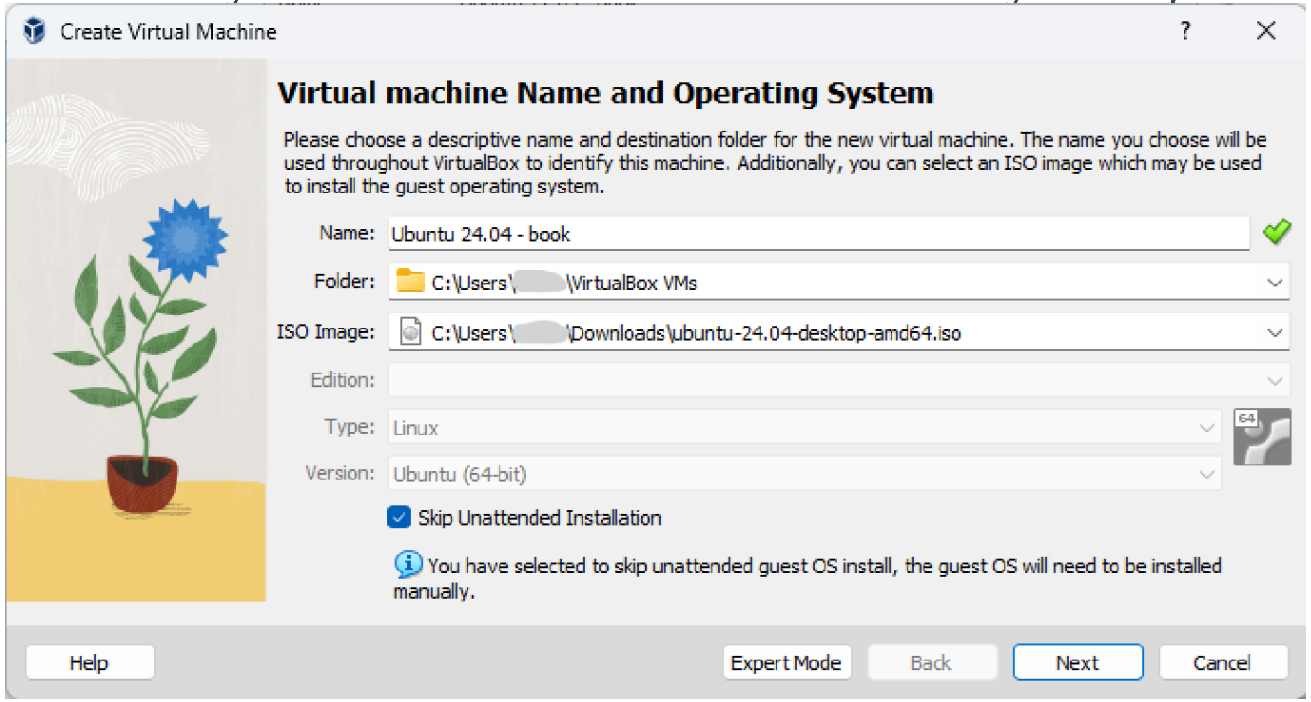
Step 3 – Creating a New VM

- Once you have installed VirtualBox and downloaded the Ubuntu .iso file, open the VirtualBox software (VirtualBox Manager) and click New.
- This will open a pop-up window where you can start to configure the new VM:

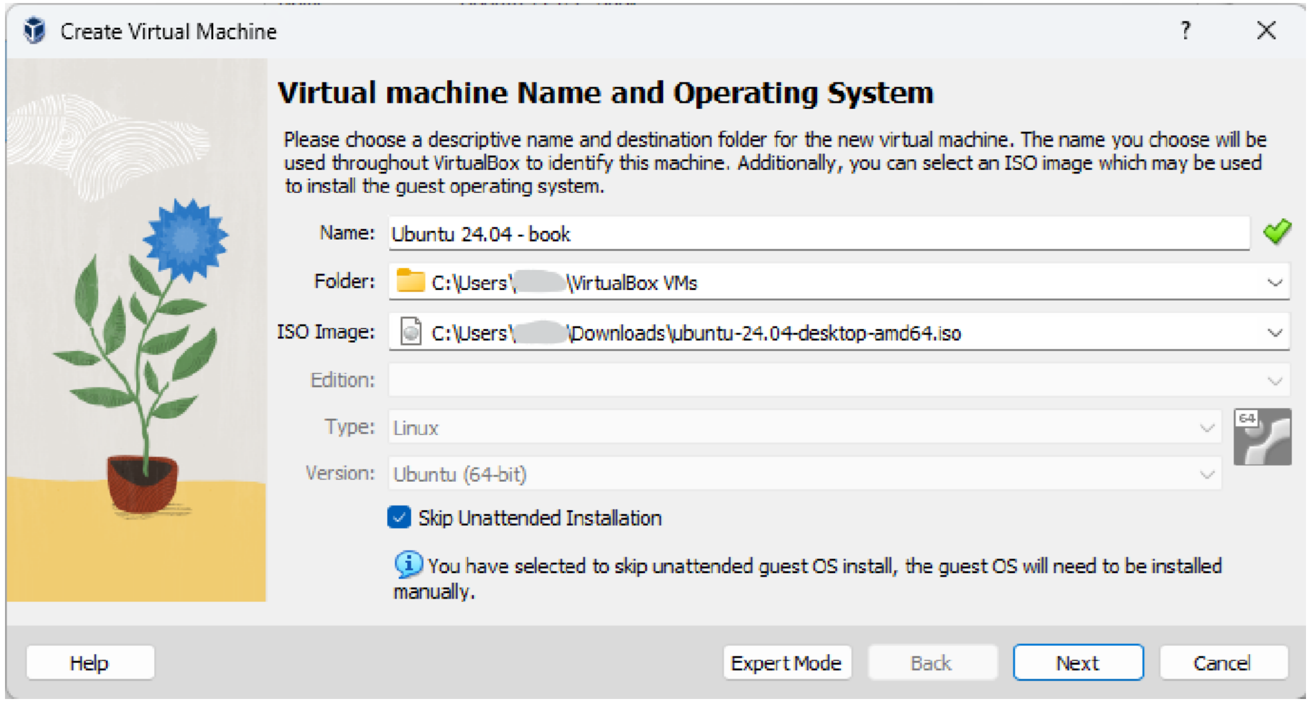


Installing Ubuntu 24.04 on a VM

- Here, we have the following values:
  - **Name:** Name the machine. This can be anything you want, just so you can recognize the machine—for example, Ubuntu 24.04 - ROS.
  - **Folder:** By default, VirtualBox will create a VirtualBox VMs folder in your user directory, where it will install all the VMs. You can keep this or change it if you want.
  - **ISO Image:** This is where you select the Ubuntu .iso file that you've just downloaded.

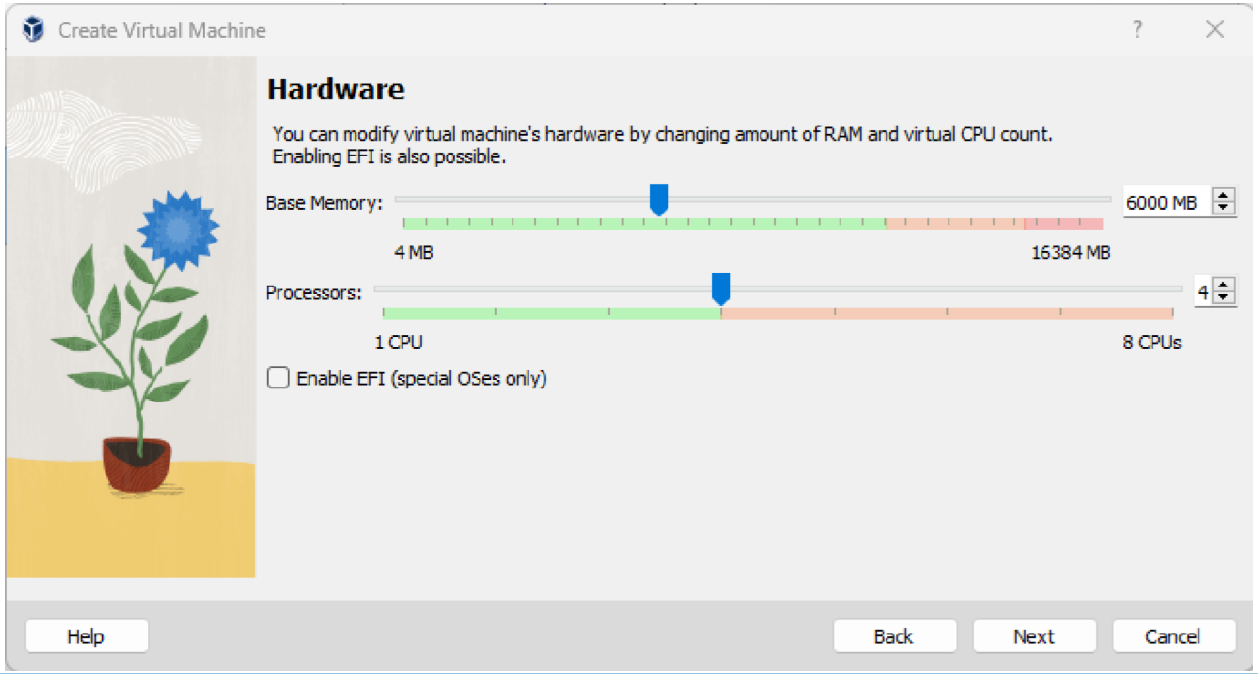


- Here, we have the following values:
  - **Type:** This should be Linux.
  - **Version:** This should be Ubuntu (64-bit).
  - **Skip Unattended Installation:** Make sure you check this box. Leaving this unchecked could be a cause of issues later.



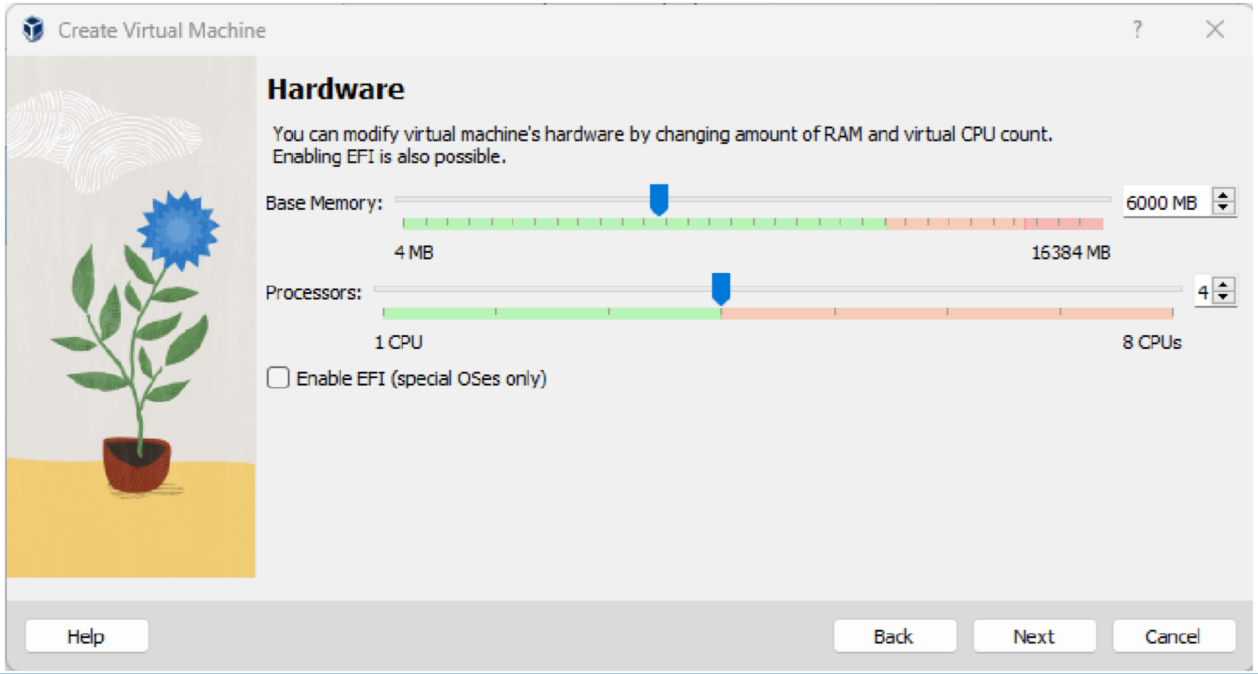
Installing Ubuntu 24.04 on a VM

- Click Next. Here, you'll need to choose how much CPU and RAM you want to allocate for the machine. This will depend on your computer's configuration.
- Here's what we recommend for the RAM allocation (Base Memory on VirtualBox):
  - If you have 16 GB or more on your computer, allocate 6 GB or a bit more; this is going to be enough.
  - If you have 8 GB RAM, allocate 4 GB.



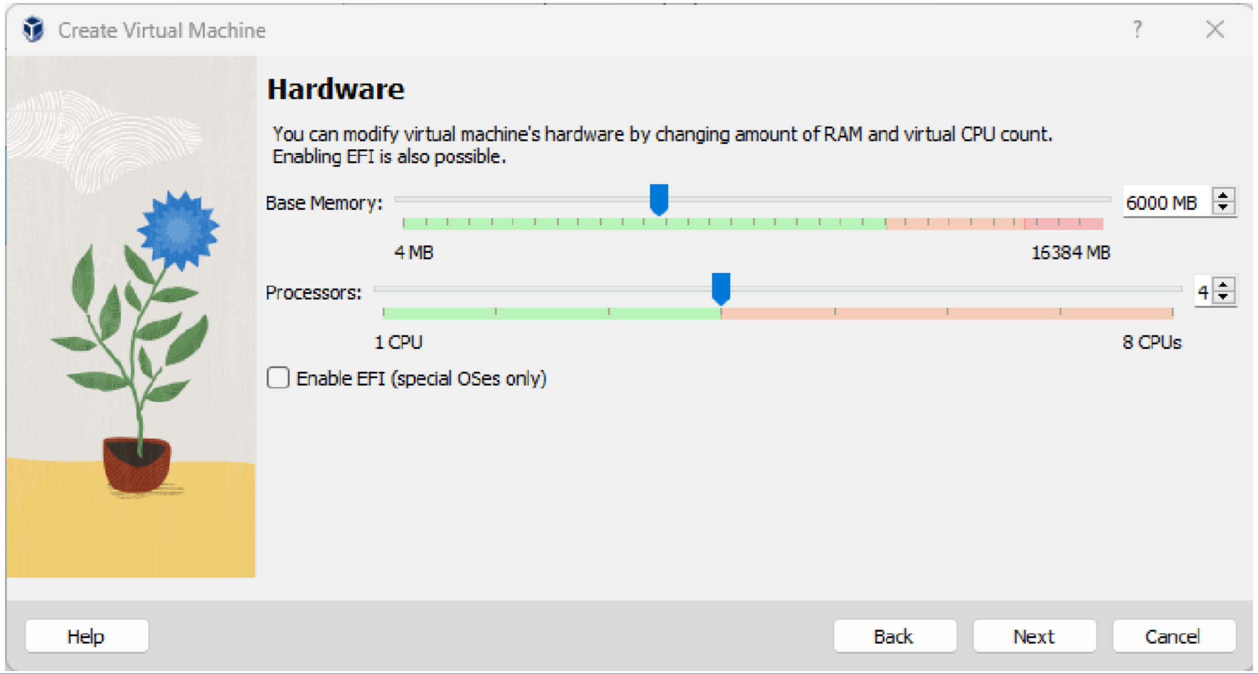
Installing Ubuntu 24.04 on a VM

- For less than 8 GB, adjust the RAM value (you can set a value now and modify it later in the settings) so that you can start the VM, open VS Code and Firefox with a few tabs, and your machine doesn’t slow down too much.
- If things get too slow, consider using a more powerful machine for learning ROS.



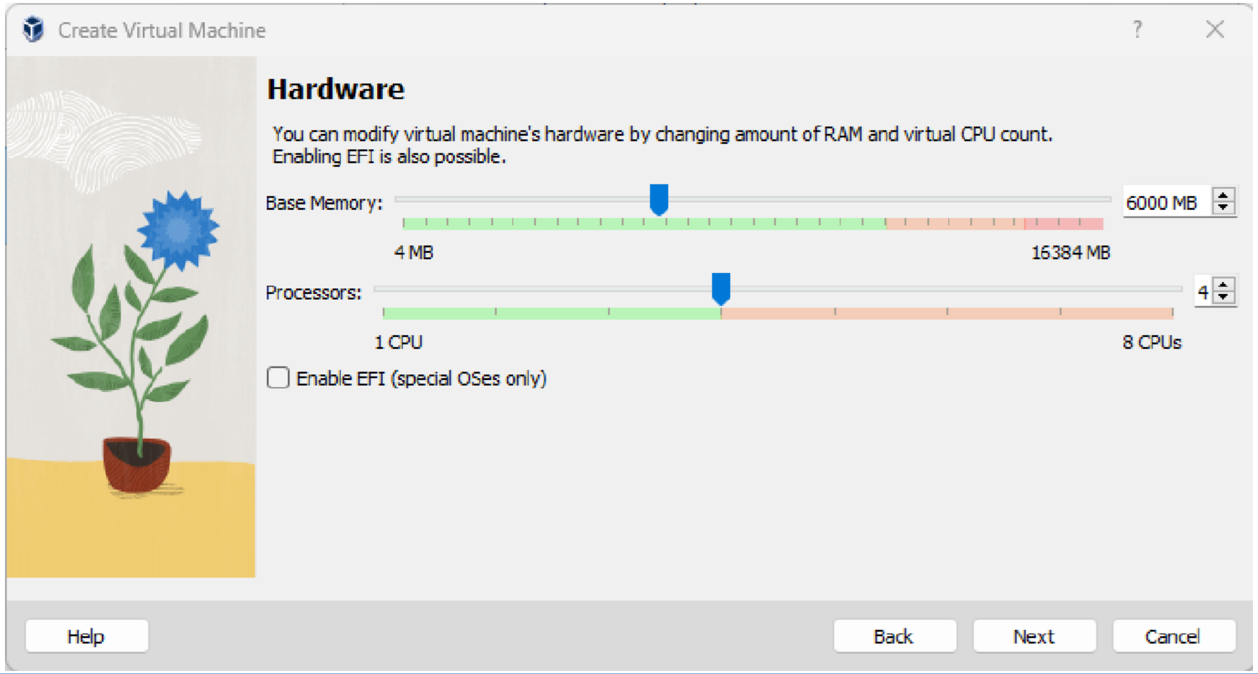
Installing Ubuntu 24.04 on a VM

- For the CPU allocation, allocate half of your CPUs.
- So, if your computer has 8 CPUs, set the value to 4. In my setup,
- Try not to go under 2 as it will be very slow with only 1 CPU.
- If in doubt, try one setting now; you can change it later.



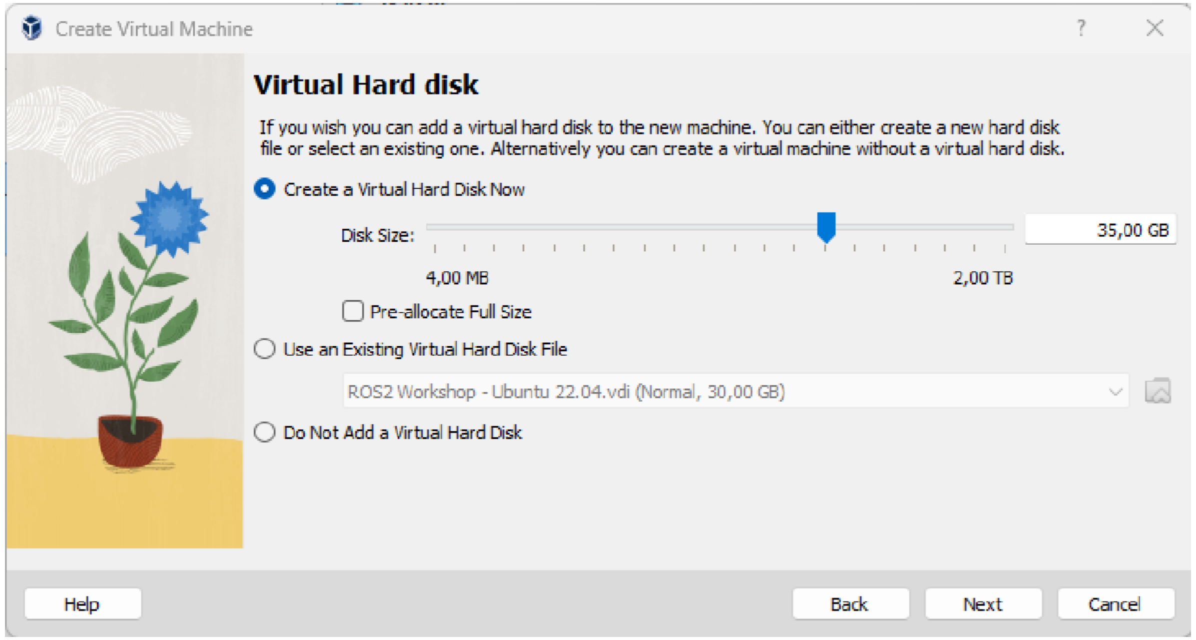
Installing Ubuntu 24.04 on a VM

- All in all, it’s better to stay in the green zone.
- If you have to push to the orange zone, then make sure that when you run your VM, you don’t run anything else on your computer (or maybe just a web browser with one tab or a PDF reader).



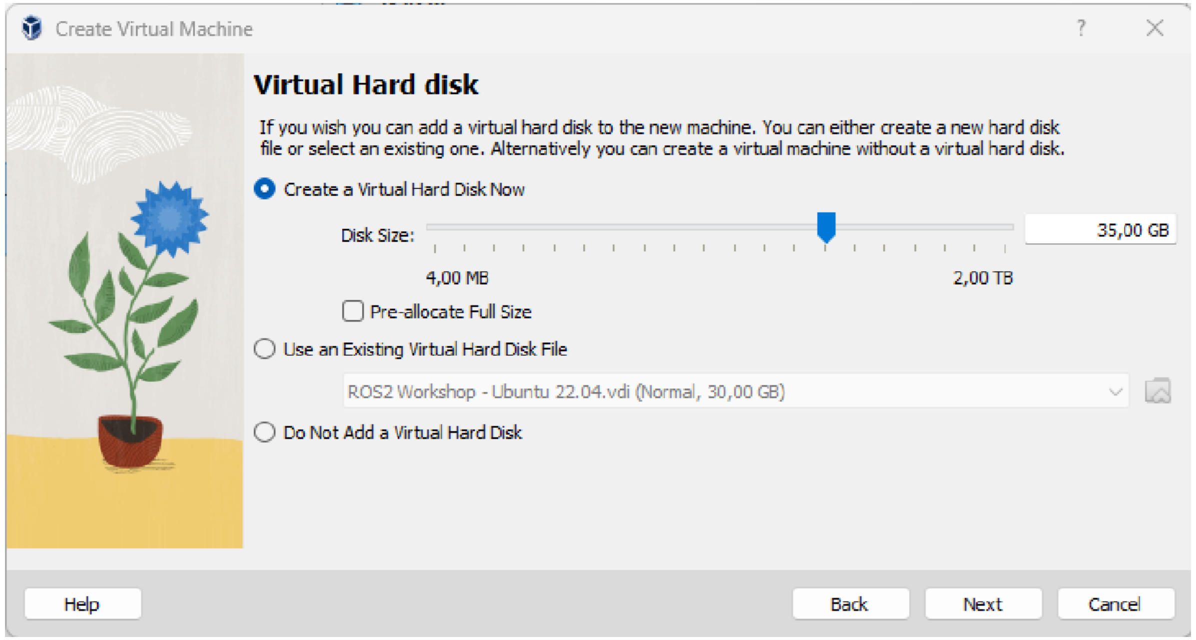
Installing Ubuntu 24.04 on a VM

- Click Next. The last thing to do for now is to configure the virtual hard disk that will be created for the VM.
- You can keep the default option (Create a Virtual Hard Disk Now). The default size is 25 GB.
- To learn ROS 2, we recommend using 30 to 40 GB minimum.



Installing Ubuntu 24.04 on a VM

- Anyway, the size of the VM will start low and expand as you install more things, so you can set a higher maximum without blocking resources.
- Keep the Pre-allocate Full Size box unchecked.



Installing Ubuntu 24.04 on a VM

- Click Next. You will now get a recap of all options you chose in the previous steps. Then, click Finish.

- You will see the newly created VM on the left in VirtualBox Manager.

- Before you start the VM, there are a few more things we need to configure.

- Open the settings for the VM (either select it and click on the Settings button or right-click on the VM and choose Settings).

Installing Ubuntu 24.04 on a VM

- Modify the following three settings:
  - System | Acceleration: Uncheck the Enable Nested Paging box
  - Display: Uncheck the Enable 3D Acceleration box (this one may already be unchecked)
  - Display: Increase Video Memory to 128 MB if possible

- With these settings, you can probably avoid unexpected behaviors and problems with the graphical interface in the VM.

Installing Ubuntu 24.04 on a VM

Step 4 – starting the VM and finishing the installation

- Now that the VM has been configured correctly, we need to start it to install Ubuntu, using the Ubuntu .iso file that we've downloaded and added to the settings.

- To start the VM, double-click on it in VirtualBox Manager, or select it and click on the Start button.

- You will get a boot menu. The first choice is Try or Install Ubuntu and it should already be selected. Press Enter.

Installing Ubuntu 24.04 on a VM

- Wait a few seconds; Ubuntu will start with the installation screen. Follow the configuration through the different windows.
  1. Choose your language. We recommend you keep English so you have the same configuration as the others.
  2. Skip the Accessibility menu, unless you need to set up a bigger font size, for example.
  3. Select your keyboard layout.
  4. Connect to the internet. To do so, choose Use wired connection.

Installing Ubuntu 24.04 on a VM

5. At this point, you might have a screen asking you to update the installer. In this case, click Update now. Once finished, click Close installer. Look for Install Ubuntu 24.04 LTS on the VM desktop and double-click on it. This will start the installation again from Step 1; repeat Steps 1 to 4.

5. When you're asked how you would like to install Ubuntu, choose Interactive installation.

5. For the applications to install with Ubuntu, go with Default selection. This will reduce the space used by the VM and you still get a web browser, as well as all the core basics—we don't need more than that to install ROS 2.

5. When asked if you want to install recommended proprietary software, choose Install third party software for graphics and Wi-Fi hardware.

Installing Ubuntu 24.04 on a VM

9. For the disk setup, select Erase disk and install Ubuntu. There's no risk here as we're erasing the empty virtual disk we've just created (if you were installing Ubuntu as a dual boot, you would have to choose another option).

9. Choose a username, computer name, and password. Keep things simple here. For example, I use ed for the user and ed-vm for the computer's name. Also, make sure that the password is typed correctly, especially if you've changed the keyboard layout previously.

9. Select your time zone.

9. On the recap menu, click Install and wait a few minutes.

9. When the installation is complete, a popup will ask you to restart. Click Restart now.

Installing Ubuntu 24.04 on a VM

14. You will see a message stating Please remove the installation medium, then press ENTER. There's no need to do anything here—just press Enter.

14. After booting, you will get another welcome screen popup. You can skip all the steps and click Finish to exit the popup.

- With that, Ubuntu has been installed.

Installing Ubuntu 24.04 on a VM

- Just to finish things properly, open a Terminal window and upgrade the existing packages (it's not because you just installed Ubuntu that all packages are up to date):

  $ sudo apt update

  $ sudo apt upgrade

- That's it for the installation. However, there's one more thing specific to VirtualBox that we must d so that the VM window works correctly.

Installing Ubuntu 24.04 on a VM

Step 5 – Guest Additions CD Image

- At this point, if you try to resize the window where the VM is running, you will see that the desktop resolution doesn't change.
- Also, if you try to copy/paste some text or code between your host and the VM, it probably won't work.
- To fix this, we need to install what's called the Guest Additions CD Image.

Installing Ubuntu 24.04 on a VM

- First, open a Terminal and run the following command (note that copy/paste doesn’t work yet, so you have to type it manually).

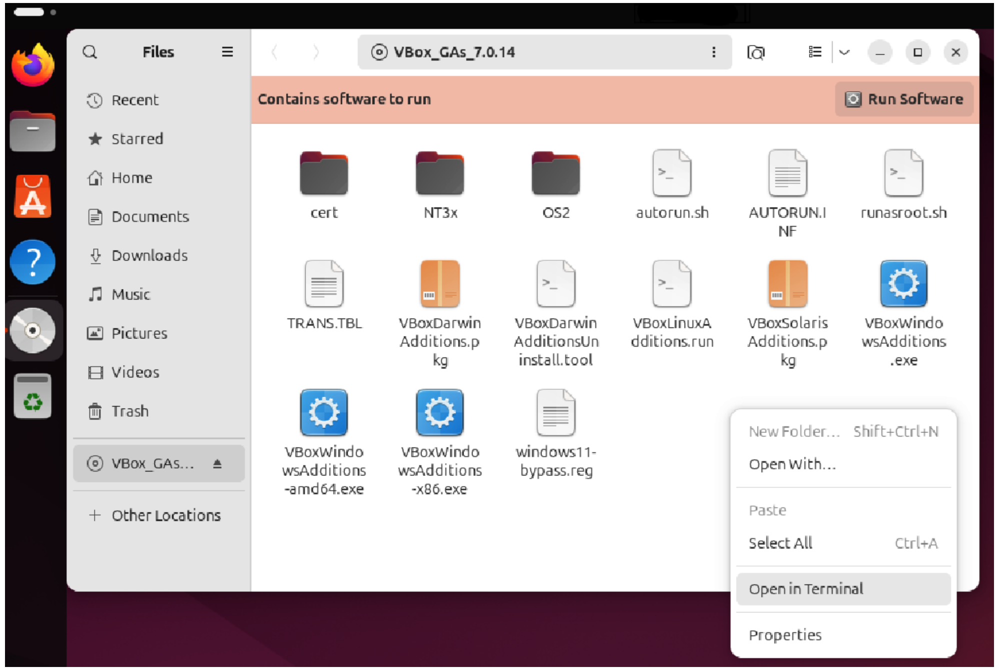
- This will install a few dependencies, all of which are required for the next step:

  sudo apt install build-essential gcc make perl dkms

- Then, on the top menu of the VM window, click Devices | Insert Guest Additions CD image.

Installing Ubuntu 24.04 on a VM

- You will see a new CD image on the Ubuntu menu (on the left). Click on it—this will open a file manager.
- Inside the file manager, right-click on the empty space and choose Open in Terminal.



Installing Ubuntu 24.04 on a VM

- You will then be inside a Terminal, where you can find a file named *VBoxLinuxAdditions.run*. You need to run this file with admin privileges:

  $ sudo ./VBoxLinuxAdditions.run

- After this, you might get a popup asking you to reboot the VM. Click Restart Now.

- If you don't get this popup, simply run sudo reboot in the Terminal.

Installing Ubuntu 24.04 on a VM

- When the VM starts again, the screen should resize automatically when you change the size of the window.

- If it doesn’t work the first time, running the command once again (sudo ./VBoxLinuxAdditions.run) and rebooting may solve the problem.

- You can then right-click on the disk and choose Eject, and you’re all set.

Installing Ubuntu 24.04 on a VM

- To enable copy/paste between the host and the VM, go to the top menu and click Devices | Shared Clipboard | Bidirectional

- Now, your VM is fully installed and configured.

Outline

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- **Installing ROS 2 on Ubuntu**
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Installing ROS 2 on Ubuntu

- Now that you have Ubuntu 24.04 installed on your computer (either as a dual boot or in a VM), let's install ROS 2.

- As discussed previously, we will install ROS Jazzy here. If you're using a different Ubuntu distribution, make sure to use the appropriate ROS 2 distribution.

- The best way to install ROS 2 is to follow the instructions on the official documentation website, for binary packages installation.

Installing ROS 2 on Ubuntu

- There are quite a few commands to run, but don’t worry—all you need to do is copy and paste them one by one.

Setting the locale

- Make sure you have a locale that supports UTF-8:

```
$ locale
```

- With this, you can check that you have UTF-8.

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- If in doubt, just run those commands one by one.

```
$ sudo apt update && sudo apt install locales
$ sudo locale-gen en_US en_US.UTF-8
$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
$ export LANG=en_US.UTF-8
```

- Now, check again; you'll see UTF-8 this time:

```
$ locale
```

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Setting up the sources

- Run those five commands one by one (note that it's better to copy/paste them from the official documentation directly):

```
$ sudo apt install software-properties-common

$ sudo add-apt-repository universe

$ sudo apt update && sudo apt install curl –y

$ sudo curl –sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg
```

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- Then add the repository to your sources list.

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

- If you are going to build ROS packages or otherwise do development, you can also install the development tools:

```
$ sudo apt update && sudo apt install ros-dev-tools
```

Installing ROS 2 on Ubuntu

- Update your apt repository caches after setting up the repositories.

```
sudo apt update
sudo apt upgrade
```

- Desktop Install (Recommended): ROS, RViz, demos, tutorials.

```
sudo apt install ros-jazzy-desktop
```

- Set up your environment by sourcing the following file

```
source /opt/ros/jazzy/setup.bash
```

Installing ROS 2 on Ubuntu

- In one terminal, source the setup file and then run a C++ talker:

```
source /opt/ros/jazzy/setup.bash
ros2 run demo_nodes_cpp talker
```

- In another terminal source the setup file and then run a Python listener:

```
source /opt/ros/jazzy/setup.bash
ros2 run demo_nodes_py listener
```

- You should see the talker saying that it’s Publishing messages and the listener saying I heard those messages. This verifies both the C++ and Python APIs are working properly. Hooray!

Setting Up Environment

- At this point, open a Terminal and run the following command:

  $ ros2

  ros2: command not found

- As we will see later, ros2 is a command-line tool we can use to run and test our programs from the Terminal.

- If this command isn't working, it means that ROS 2 hasn't been set up correctly.

Setting Up Environment

- Even if ROS 2 is installed, there's one more thing you need to do in every new session (or Terminal) where you want to use ROS 2: you need to source it in the environment.
- To do that, source this bash script from where ROS 2 is installed:

```
$ source /opt/ros/jazzy/setup.bash
```

- After you run this, try executing the ros2 command again. This time, you should get a different message (usage message).
- This means that ROS 2 is correctly installed and set up in your environment.

Setting Up Environment

- You must source this bash script every time you open a new session or Terminal.
- To make things easier and so you don't forget about it, let's just add this command line to the .bashrc file.
- If you don't know what .bashrc is, simply put, it's a bash script that will run every time you open a new session (that can be an SSH session, a new Terminal window, and so on).

Setting Up Environment

- This .bashrc file is specific to each user, so you will find it in your home directory (as a hidden file because of the leading dot).

- You can add the source line to the .bashrc file with this command:

$ echo 'source /opt/ros/jazzy/setup.bash' >> ~/.bashrc

- You could also just open the .bashrc file directly with any text editor—gedit, nano, or Vim—and add the source line at the end. Make sure it’s only added once.

Setting Up Environment

- Once you've done this, any time you open a new Terminal, you can be sure that this Terminal is correctly sourced, and thus you can use ROS 2 in it.

- Now, to make a final check, open a Terminal and run this command (there's no need to understand anything for now; it's just to verify the installation):

  ros2 run turtlesim turtlesim_node

- This will print a few logs, and you should see a new window with a turtle in the middle.

Setting Up Environment

- To stop the program, press Ctrl + C in the Terminal where you ran the command.

- That's it for installing and configuring ROS 2.

Outline

- Which ROS 2 Distribution to Choose?
- Installing the OS (Ubuntu)
- Installing Ubuntu 24.04 on a VM
- Installing ROS 2 on Ubuntu
- **Installing Tools**

Installing VS Code

- Visual Studio Code (VS Code) is a quite popular IDE used by many developers. What makes it nice for us is its good support for ROS development.

- VS Code is free to use and open source; you can even find its code on GitHub. To install VS Code, open a Terminal and run the following command:

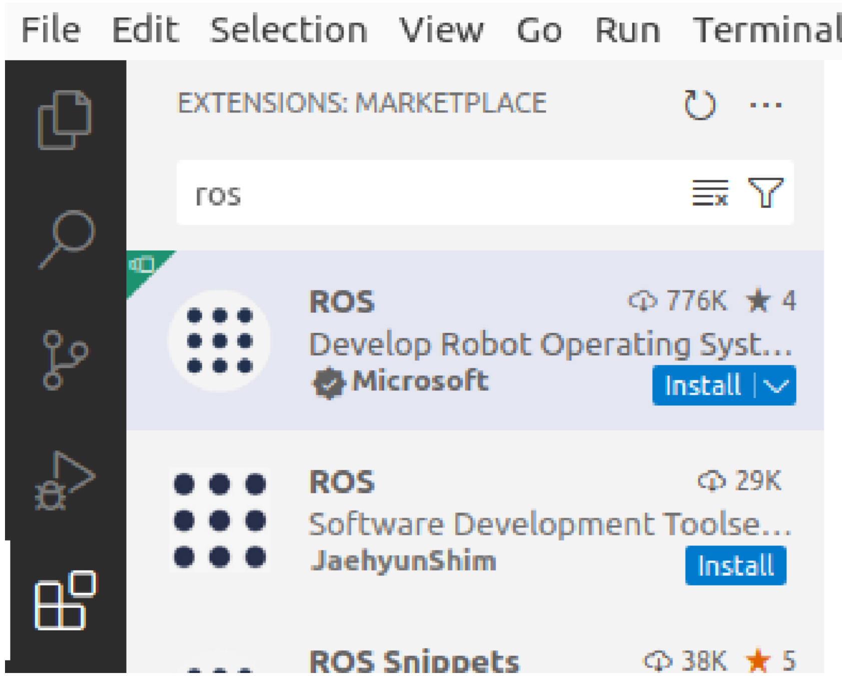
```
sudo snap install code --classic
```

Installing VS Code

- After installing it, you can open VS Code by searching for it in the Ubuntu applications, or simply by running code in the Terminal.

- Now, start VS Code and go to the Extensions panel—you can find it on the left menu.

- There, you can search for the ROS extension by typing ros. There are quite a few; choose the one developed by Microsoft. This extension is compatible with both ROS 1 and ROS 2, so there's no problem here.



Installing VS Code

- Install this extension. This will also install a bunch of other extensions, notably Python and C++ extensions, which are quite useful when writing code.

- On top of this, also install the Cmake extension by twxs (just type cmake and you'll find it).

- With this, we get nice syntax highlighting when writing into CMakeLists.txt files, which is something we will have to do quite often with ROS 2.

Installing Terminator

- As you develop with ROS 2, you will often need to open several Terminals: one for compiling and installing, a few to run the different programs of your application, and a few more for introspection and debugging.

- It can become quite difficult to keep track of all the Terminals you use, so as best practice, it's nice to have a tool that can easily handle multiple Terminals in one window.

Installing Terminator

- There are quite a few tools for doing this. The one that we are going to use is called terminator.

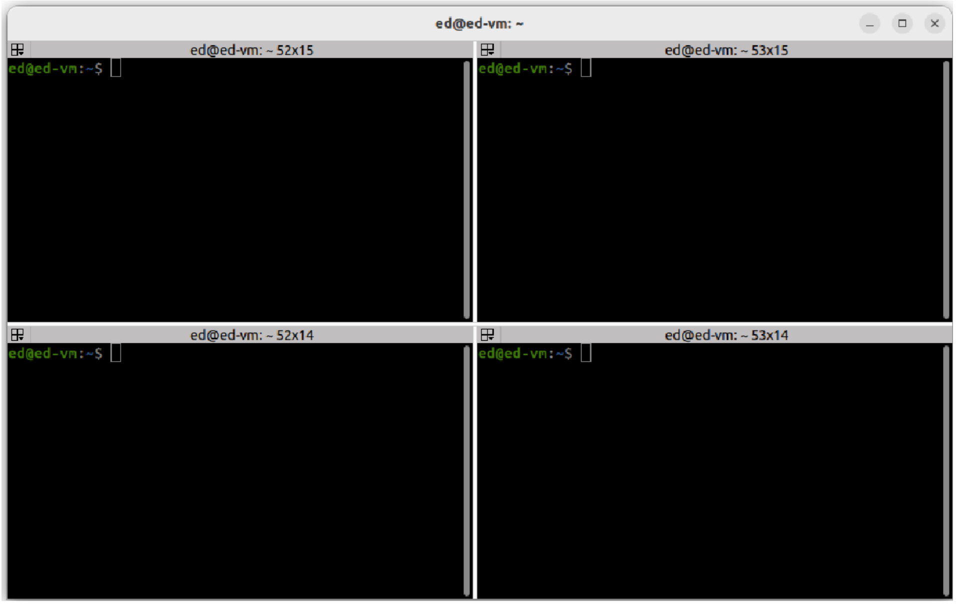
- To install terminator, run the following command:

  sudo apt install terminator

- Then, you can find it from the applications menu, run it, right-click on the left bar menu, and choose Pin to Dash so that it stays there and becomes easy to start.

Installing Terminator

- You can find all the commands for terminator online, but here are the most important ones to get started:
    - Ctrl + Shift + O: split the selected Terminal horizontally.
    - Ctrl + Shift + E: split the selected Terminal vertically.
    - Ctrl + Shift + X: make the current Terminal fill the entire window. Use again to revert.
    - Ctrl + Shift + W: close a Terminal.



Installing Terminator

- Whenever you split a Terminal, this Terminal becomes two different Terminals, each being one session.

- Thus, you can easily split your Terminal into four or six; this will be enough to run most of your ROS 2 applications.

- Since we previously added the line to source ROS 2 in the .bashrc file, you can use ROS 2 directly in each new Terminal.

Thank you for your attention.

Any Questions?